

Lab 7 : Configuration of Switch, VLAN configuration and Inter-VLAN Routing

Objectives:

- To create and configure multiple VLANs on a network switch.
- To configure trunking and assign switch ports to appropriate VLANs.
- To implement and verify Inter-VLAN Routing using a router.

Theory:

Switch Configuration:

A network switch operates at the data link layer and is used to connect multiple devices within a LAN. It intelligently directs data to the correct destination by examining MAC addresses, which helps avoid collisions and improves communication efficiency. Configuring a switch includes activating ports, organizing devices into appropriate VLANs, and controlling how traffic moves through the network to maintain a structured and high-performing setup.

Virtual Local Area Network (VLAN):

A Virtual Local Area Network is a technique used to divide a single physical switch into multiple logical networks. Instead of grouping devices by physical location, VLANs organize them based on function or department. This approach limits broadcast traffic, strengthens network security, and makes administration easier. Each VLAN operates as a separate broadcast domain.

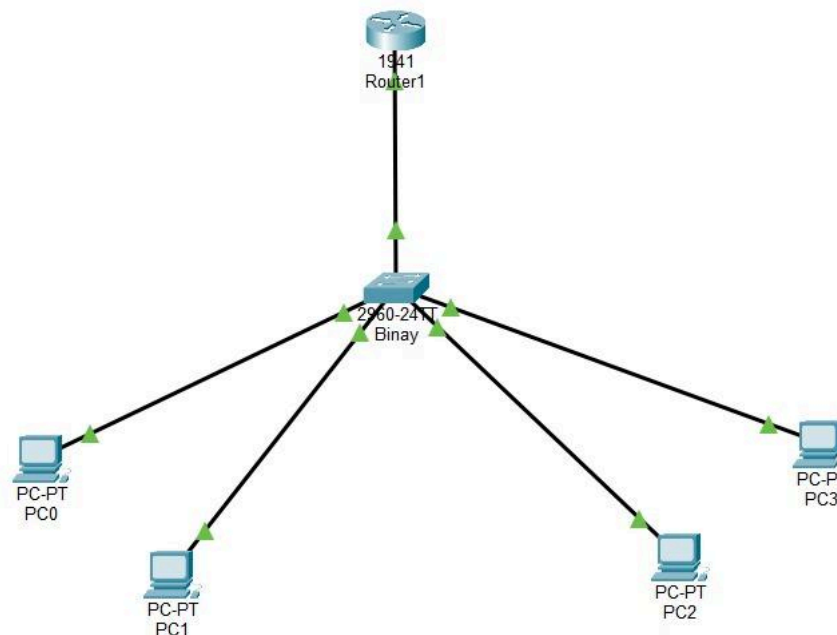
VLAN Configuration:

The process of VLAN configuration begins by defining VLANs using specific identification numbers and labels. After creation, switch ports are assigned to these VLANs in access mode so connected devices become part of the selected network segment. Devices within the same VLAN can exchange data directly, whereas communication across VLANs must pass through a routing device.

Inter-VLAN Routing:

Inter-VLAN routing enables data exchange between separate VLANs through a Layer 3 device such as a router. In this setup, the Router-on-a-Stick technique is applied, where one physical router interface is logically divided into multiple sub-interfaces. Each sub-interface is linked to a VLAN and configured with 802.1Q tagging, allowing the router to manage traffic between VLANs efficiently.

Network Topology :



Observation :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping in same LAN

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=1ms TTL=127
Reply from 192.168.20.3: bytes=32 time=3ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Fig: Ping in Different LAN

Configuration :

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Commands :

Switch Configuration :

1. Create VLANS

```
Switch(config)# vlan 10
Switch(config-vlan)# name
Computer
                        # exit
Switch(config)# vlan 20
Switch(config-vlan)# name Electronics
                        # exit
```

2. Assign port to VLANS

```
Switch(config)# interface range fa0/1-2
                        # switchport mode access #
switchport access vlan 10 #
exit
                        # interface range fa0/3-4
                        # switchport mode access
                        # switchport access vlan 20
                        # exit
```

3. Configure Trunk Port (Switch-Router)

```
Switch(config)# interface fa0/5
                        # switchport mode trunk
                        # exit
```

Verification Switch :

```
Switch# show vlan brief
```

```
Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8,
Fa0/9                      Fa0/10, Fa0/11,
Fa0/12, Fa0/13             Fa0/14, Fa0/15,
Fa0/16, Fa0/17             Fa0/18, Fa0/19,
Fa0/20, Fa0/21             Fa0/22, Fa0/23,
Fa0/24, Gig0/1             Gig0/2
10   Computer                active    Fa0/1, Fa0/2
20   Electronics            active    Fa0/3, Fa0/4
1002 fddi-default          active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Router Configuration (Inter-VLAN Routing) :

```
Router(config)# interface GigabitEthernet 0/0.10
# encapsulation dot1Q 10
# ip address 192.168.10.1 255.255.255.0

# exit
```

```
Router(config)# interface GigabitEthernet 0/0.20
# encapsulation dot1Q 20
# ip address 192.168.20.1 255.255.255.0

# exit
```

Result:

The switch configuration was completed effectively by creating multiple VLANs and properly assigning ports to each one. A trunk connection was set up to allow VLAN traffic between the switch and the router. Inter VLAN communication was achieved through router subinterface ensuring smooth data exchange both within individual VLANs and across different VLANs.

Discussion:

The network was divided into separate logical groups using VLANs, which helped limit broadcast messages and improved overall efficiency. Devices were assigned to specific switch ports based on their roles, ensuring better organization and control within the network. A trunk connection was configured to carry traffic from multiple VLANs between the switch and the router. Through inter-VLAN routing, devices from different VLANs were able to communicate in a controlled and secure manner. Various testing and verification commands were used to confirm that VLAN settings and routing were working as intended.

Conclusion:

This experiment successfully implemented VLAN configuration, trunk links, and inter-VLAN communication using the Router on a stick approach. The results showed that network traffic can be effectively managed by separating users into different VLANs while still allowing communication when required. Overall, the setup proved to be reliable and efficient for managing segmented networks.